

RESEARCH ARTICLE

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Scalable Multimodal Neural Architecture for Real-Time Macromolecular Structure Prediction

High-Throughput Geometric Deep Learning with Integrated Physical Energy Constraints

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ABSTRACT

Macromolecular structure determination remains a computational bottleneck in drug discovery and synthetic biology. Here we introduce **Synapse-Fold**, an end-to-end multimodal geometric neural network capable of predicting tertiary protein-ligand complexes at sub-angstrom resolution in real time. By integrating equivariant graph convolutions with differentiable thermodynamic boundary constraints, Synapse-Fold achieves a 140-fold speedup compared to state-of-the-art conformational sampling routines while reducing prediction error by 38% on benchmark datasets (Nexa-PDB2025). The architecture incorporates an adaptive SE(3)-equivariant attention layer that natively models side-chain dynamics and non-covalent interactions under flexible cellular microenvironments. Validation across 4,200 diverse protein targets synthesized at the Meridian Bio-Foundry confirms that predicted structural dynamics closely match cryo-EM density maps (mean RMSD of 0.84 Å). This hybrid deep learning and physical modeling paradigm provides a foundation for high-throughput automated biotherapeutic design.

Keywords: Geometric Deep Learning • Macromolecular Structure • Equivariant Graph Neural Networks • Cryo-EM Validation • Synapse Engine

1 Introduction

Determining three-dimensional macromolecular structures and understanding their conformational landscapes are central challenges in modern structural biology and rational drug design [1, 2]. Although recent advances in deep learning—most notably transformer-based homology prediction models—have dramatically improved single-chain structure prediction accuracy, accurately modeling multi-component protein-ligand interactions under physiological conditions remains challenging. Traditional molecular dynamics (MD) simulations provide atomic trajectories with high physical fidelity, but their intense computational requirements prevent their deployment in large-scale virtual screening campaigns.

To resolve this trade-off between physical rigor and computational speed, we present **Synapse-Fold**, a multimodal geometric deep learning framework. Unlike conventional models that treat structural prediction as an isolated geometric translation task, Synapse-Fold explicitly encodes physical energy surfaces within its latent representations. By embedding SE(3)-equivariant graph neural network (GNN) layers with differentiable molecular mechanics loss terms, our system enforces bond length constraints, steric overlap penalties, and electrostatic potential boundaries directly during the forward pass.

Key Highlights

- Sub-angstrom resolution (0.84 Å RMSD) across 4,200 target complexes.
- 140× inference speedup over iterative Monte Carlo docking pipelines.
- Seamless integration of SE(3) equivariance with physical force-field priors.
- Validated experimentally on the Apex-9 benchmark dataset.

Open Data & Code
All model weights, dataset manifests (Nexa-PDB2025), and evaluation scripts are available under an open license at: github.com/vertex-lab/synapse-fold